APCSA FRQ High Score Tips

Using Scoring Guidelines • Dice Class Example • Matrix (2D Array) Example

FRQ Scoring Guidelines Overview

- Partial credit is KEY — rubrics give points for every correct piece
- 1 pt: class header
- 1–2 pts: instance variables (private + correct type)
- 1–2 pts: constructor (parameters & initialization)
- 1 pt each: method headers + logic
- Arrays / ArrayLists: correct bounds, counters, and return values

1. Class Problems — High-Score Checklist

- Class header: public class Name {}
- Instance variables: private Type varName;
- Constructor: public Name(params) { var = param; }
- Accessor (getter): public Type getVar() { return var; } — no state change
- Mutator (modifier): public void setVar(Type v) { var = v; } — changes state
- Always declare instance variables private (easy encapsulation point!)

Dice Example — Complete Class (FRQ Style)

```java
public class Die {
    // Instance variables
    private int numSides;

    // Constructor
    public Die(int sides) {
        numSides = sides;
    }

    // Accessor (getter)
    public int getNumSides() {
        return numSides;
    }

    // Mutator (modifier)
    public void setNumSides(int sides) {
        if (sides > 0) numSides = sides;
    }

    // Common method
    public int roll() {
        return (int)(Math.random() * numSides) + 1;
    }
}
```

2. Array & ArrayList — High-Score Tips

- Initialize counter BEFORE the loop: int count = 0;
- Correct iteration boundary: for(int i = 0; i < arr.length; i++)
- ArrayList uses .size() — NOT .length
- Always return the correct variable (count / sum / result)
- Off-by-one errors are the #1 point killer in FRQs

Matrix (2D Array) Example — Correct Implementation

```
int[][] matrix = {
   {1, 2, 3},
   {4, 5, 6},
   {7, 8, 9}
};

// Correct 2D iteration (max points!)
int sum = 0;
for (int r = 0; r < matrix.length; r++) {          // rows
   for (int c = 0; c < matrix[r].length; c++) {   // columns
      sum += matrix[r][c];
   }
}
return sum;   // ← Return the correct accumulated variable!
```

Extra High-Score Insights & Best Practices

- Use enhanced for-loop when index is not needed: for(int val : arr)
- Test mentally with small data (e.g. 6-sided die, 3×3 matrix)
- Exact method names and return types matter — read FRQ carefully
- Practice past FRQs with official College Board rubrics
- Partial credit on each part is enough to score 4 or 5!